

Hanseul Cho (조한슬)

PHD CANDIDATE AT KAIST AI. EX-INTERN AT GOOGLE NYC. WORKING ON OPT/ML THEORY & LLMs.

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Personal Profile

I am a Ph.D. candidate at Kim Jaechul Graduate School of AI of Korea Advanced Institute of Science and Technology (KAIST AI). I am fortunate to be advised by Prof. Chulhee “Charlie” Yun of Optimization & Machine Learning (OptiML) Laboratory, KAIST AI. Previously, I worked at Google NYC as an intern (Student Researcher), hosted by Srdinadh Bhojanapalli. Also, I completed my M.Sc. (in AI) and B.Sc. (in Math, minor in CS, *Summa Cum Laude*) at KAIST.

My primary research interests lie in optimization, machine learning (ML), and deep learning (DL). During my journey to a Ph.D., my ultimate research goal is to **rigorously understand and practically overcome** the following **three critical challenges in ML/DL**:

Generalizability 🌱

Generalization capabilities of modern language models.

(e.g., length generalization and compositional generalization of Transformers)

Adaptability 🌱

Training adaptable models under an evolving environment.

(e.g., continual learning, maintaining the plasticity of neural networks, sample-efficient reinforcement learning)

Multifacetedness 🌱

Learning with multiple (possibly conflicting and/or orthogonal) goals.

(e.g., minimax optimization, bi-level optimization, fairness in ML)

Publications

International Conferences

- [C10] Xu, Zhiwei[☆], Shihao Wu[☆], **Hanseul Cho**, Wei Hu, Yixin Wang. Data-Constrained Language Model Pretraining: Improved Regularization and Scaling Laws. In *Advances in Neural Information Processing Systems (NeurIPS)*. 2026. [arXiv] [OpenReview]
• Short version in *ICML 2026 Workshop on Resource-Adaptive Foundation Model Inference (AdaptFM)*.
- [C9] Chang, Hoyeon[☆], Jinho Park[☆], **Hanseul Cho**[☆], Sohee Yang, Miyoung Ko, Hyeonbin Hwang, Seungpil Won, Dohaeng Lee, Youbin Ahn, Minjoon Seo. Characterizing Pattern Matching and Its Limits on Compositional Task Structures. In *Proceedings of the International Conference on Learning Representations (ICLR)*. 2026. [arXiv] [OpenReview]
- [C8] **Cho, Hanseul**[☆], Jaeyoung Cha[☆], Srinadh Bhojanapalli, Chulhee Yun. Arithmetic Transformers Can Length-Generalize in Both Operand Length and Count. In *Proceedings of the International Conference on Learning Representations (ICLR)*. 2025. [arXiv] [OpenReview]
- [C7] Jung, Hyunji[☆], **Hanseul Cho**[☆], Chulhee Yun. Convergence and Implicit Bias of Gradient Descent on Continual Linear Classification. In *Proceedings of the International Conference on Learning Representations (ICLR)*. 2025. [arXiv] [OpenReview]
- [C6] **Cho, Hanseul**[☆], Jaeyoung Cha[☆], Pranjali Awasthi, Srinadh Bhojanapalli, Anupam Gupta, Chulhee Yun. Position Coupling: Improving Length Generalization of Arithmetic Transformers Using Task Structure. In *Advances in Neural Information Processing Systems (NeurIPS)*. 2024. [arXiv] [OpenReview]
• Short version in *ICML 2024 Workshop on Long-Context Foundation Models (LCFM)*.
- [C5] Shin, Baekrok[☆], Junsoo Oh[☆], **Hanseul Cho**, Chulhee Yun. DASH: Warm-Starting Neural Network Training in Stationary Settings without Loss of Plasticity. In *Advances in Neural Information Processing Systems (NeurIPS)*. 2024. [arXiv] [OpenReview]
• Short version in *ICML 2024 Workshop on Advancing Neural Network Training (WANT): Computational Efficiency, Scalability, and Resource Optimization*.
- [C4] Lee, Jaewook[☆], **Hanseul Cho**[☆], Chulhee Yun. Fundamental Benefit of Alternating Updates in Minimax Optimization. In *Proceedings of the International Conference on Machine Learning (ICML)*. 2024. [arXiv] [OpenReview]
• **Spotlight at ICML 2024. (Top 3.5%: (144+191) of 9,473 valid submissions)**
• Short version in *ICLR 2024 Workshop on Bridging the Gap Between Practice and Theory in Deep Learning (BGPT)*.
- [C3] Lee, Junghyun[☆], **Hanseul Cho**[☆], Se-Young Yun, Chulhee Yun. Fair Streaming Principal Component Analysis: Statistical and Algorithmic Viewpoint. In *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. [arXiv] [OpenReview]
- [C2] Lee, Hojoon[☆], **Hanseul Cho**[☆], Hyunseung Kim[☆], Daehoon Gwak, Joonkee Kim, Jaegul Choo, Se-Young Yun, Chulhee Yun. PLASTIC: Improving Input and Label Plasticity for Sample Efficient Reinforcement Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. [arXiv] [OpenReview]
- [C1] **Cho, Hanseul**[☆] and Chulhee Yun. SGDA with Shuffling: Faster Convergence for Nonconvex-PL Minimax Optimization. In *Proceedings of the International Conference on Learning Representations (ICLR)*. 2023. [arXiv] [OpenReview]

[☆] Co-first authors: These authors contributed equally.

[★] Sole first authors.

International Workshop

- Jung, Minchan[☆], **Hanseul Cho**[☆], Chulhee Yun. Stronger Lower Bounds for (Non-)Anytime Acceleration of Gradient Descent. In *NeurIPS 2026 Workshop on Optimization for Machine Learning*. 2026. [arXiv] [OpenReview]
- Cha, Jaeyoung[☆], **Hanseul Cho**[☆], Chulhee Yun. Layer Verification Accelerates Speculative Tree Decoding. In *ICML 2026 Workshop on Resource-Adaptive Foundation Model Inference (AdaptFM)*. 2026. [OpenReview]

Notable Domestic Journals/Conferences

- **Cho, Hanseul**[☆], Baekrok Shin[☆], Chaewon Moon[☆], Sang-Geun Hong, U-Ju Byeon, Jin-Yong Sung, Chulhee Yun. Deep Model-Based Optimization of Jamming Effectiveness under Aircraft AESA Radar Operational Environments. *The Journal of Korean Institute of Communications and Information Sciences (J-KICS)*, vol. 50, no. 11, pp. 1647-1659, 2025. DOI: 10.7840/kics.2025.50.11.1647. [Info]
- Jung, Hyunji[☆], **Hanseul Cho**[☆], Chulhee Yun. Convergence and Implicit Bias of Gradient Descent on Continual Linear Classification. *Short version in the 11th Joint Conference of Korean Artificial Intelligence Association (JKAIA 2024)*.
 - **Best Paper Award (Top 3) & Oral presentation.**
- **Cho, Hanseul**[☆] and Chulhee Yun. SGDA with Shuffling: Faster Convergence for Nonconvex-PL Minimax Optimization. *Short version in the 7th Joint Conference of Korea Artificial Intelligence Association (JKAIA 2022)*.
 - **NAVER Outstanding Theory Paper Award (Top 3) & Oral presentation.**

Education

Korea Advanced Institute of Science and Technology (KAIST)

Seoul, South Korea

Ph.D. in Artificial Intelligence

Sept. 2023 – Current

- Advisor: Prof. **Chulhee Yun** (Optimization & Machine Learning (**OptiML**) Laboratory, Kim Jaechul Graduate School of AI (GSAI), KAIST)
- Anticipated Graduation Date: Between Aug 2027 & Feb 2028

KAIST

Seoul, South Korea

M.Sc. in Artificial Intelligence

Mar. 2022 – Aug. 2023

- Advisor: Prof. **Chulhee Yun** (Optimization & Machine Learning (**OptiML**) Laboratory, Kim Jaechul Graduate School of AI (GSAI), KAIST)
- Thesis: “Improved Convergence Rate of SGDA by Shuffling: Focusing on the Nonconvex-PL Minimax Problems” (Approved by Prof.’s **Chulhee Yun**, **Se-Young Yun**, & **Donghwan Kim**)
- GPA: 4.22/4.3

KAIST

Daejeon, South Korea

B.Sc. in Mathematical Sciences

Mar. 2017 – Feb. 2022

- Minor in Computing Sciences
- Summa Cum Laude (GPA: 4.05/4.3)

University of Twente

Enschede, Netherlands

Exchange Student Program

Feb. 2020 – Jul. 2020

- Major in Applied Mathematics

Incheon Science High School (ISHS)

Incheon, South Korea

High School

Mar. 2015 – Feb. 2017

- Early graduation by one year (i.e., two-year course)

Experiences

Google

New York, NY, United States

Internship: Student Researcher Program (On-Site), Engineering

May 5th 2025 – Aug. 22nd 2025

- Host: **Srinadh Bhojanapalli** (Staff Research Scientist at Google DeepMind)
- Notable Co-workers/-advisors: **Hrayr Harutyunyan** & **Amir Keivan Mohtashami** (Research Scientists at Google DeepMind)
- Office: Google NYC, 9th Building (111 8th Ave, New York, NY)
- Research Topic: Advanced attention mechanisms of Transformers for long contexts

KAIST 2021 Post-AI Research Project

Daejeon, South Korea

Undergraduate Researcher

May 2021 – Dec. 2021

- Jointly advised by Prof. **Sangyoon Yi** (DS Lab, GSFS, KAIST) & Prof. **Jinkyoo Park** (Sys. Int. Lab, ISysE, KAIST)
- Project: Research on ‘AI-augmented Organizations’ of Collaborative Decision Making and Learning. My contributions are as follows:
 1. *Algorithm Design*: Devised a model-based randomized algorithm for a single-player finite-horizon NK landscape optimization game
 2. *Experiment Assistant*: Conducted human-subject experiments on human-AI cooperation based on the algorithm that I devised

Individual Study: Optimization for Deep Learning

Undergraduate Student @ KAIST

- Advised by Prof. Jinwoo Shin (ALIN Lab, GSAI, KAIST)
- (1) Gradient-based optimizers for large-batch setting (e.g., LARS & LAMB); (2) Theoretical analysis on gradient clipping (paper reading)

Daejeon, South Korea

Mar. 2021 – Jun. 2021

Individual Study: Deep Learning in Computer Vision

Undergraduate Student @ KAIST

- Advised by Prof. Jong-chul Ye (BISPL, BBE, KAIST)
- Assignment: Semantic segmentation of kidney tumor with U-Net (with KiTS19 challenge dataset)
- Self-taught PyTorch coding on Linux Ubuntu

Daejeon, South Korea

Sep. 2020 – Feb. 2021

Individual Study: Statistical Learning Theory

Undergraduate Student @ KAIST

- Advised by Prof. Yeonseung Chung (MAS, KAIST)
- Resource: Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. "An Introduction to Statistical Learning: with Applications in R." Springer, 2013. [\[link\]](#)

Daejeon, South Korea

Jun. 2020 – Aug. 2021

Seminars

LeT (Learning Theory) Workshop

Organizing Committee

- Homepage: let-workshop.github.io/
- Through this workshop, we aim to bring together researchers interested in ML/AI theory in the Republic of Korea.
- Other members of the organizing committee: Junghyun Lee (KAIST), Dongyeop Lee (POSTECH), Prof. Chulhee Yun (KAIST), Prof. Kwang-Sung Jun (POSTECH), and Prof. Namhoon Lee (POSTECH).

Pohang, South Korea

Oct. 7th–8th, 2026

SNU-KAIST AI/ML Theory Workshop

Organizer

- Homepage: nick-jhlee.github.io/snu-kaist-workshop
- Jointly organized by three research groups of Prof. Ernest K. Ryu, Prof. Min-hwan Oh, and Prof. Chulhee Yun.

Gangneung, South Korea

Aug. 12th–14th, 2024

Machine/Deep Learning Theory + Physics (MDLTP) Seminar

Organizer

- Homepage: sites.google.com/view/mdlt-p
- Jointly organized by OSI Lab, OptiML, and CSSPL
- Topics: Learning theory, loss landscape, trajectory analysis, (stochastic) optimization, high-dimensional statistics, statistical/mathematical physics, scientific machine learning, and more.

Seoul, South Korea

Jul. 2022 – Feb. 2023

Geometric Deep Learning Seminar

Participant

- A seminar organized by OptiML and OSI Lab
- Resources: [\[Homepage\]](#) [\[Lecture Videos\]](#) [\[Book\]](#)

Seoul, South Korea

2022

Awards

2026	Gold Reviewer (Top 25%: 4,439 of 17,749 reviewers) , ICML 2026	Seoul, South Korea
2025	Top Reviewer (Top 1.88%: 206 of 10,943 reviewers) , ICML 2025	Vancouver, Canada
2024	Best Paper Award (Top 3) , JKAIA 2024	South Korea
2024	Top Reviewer (Top 8.60%: 1,304 of 15,160 reviewers) , NeurIPS 2024	Vancouver, Canada
2022	NAVER Outstanding Theory Paper Award (Top 3) , JKAIA 2022	South Korea
2022	Summa Cum Laude , Bachelor's, KAIST	South Korea
2017 – 2020	The National Scholarship for Science and Engineering , Korea Student Aid Foundation	South Korea
2017 Fall	Dean's List , The School of Freshman, KAIST	South Korea

Services

Top-tier ML Conference/Journal Reviewer (35 papers)

From time to time

- **NeurIPS**: 2023 (🏆🏆), 2024* (🏆🏆🏆🏆🏆), 2025 (🏆), 2026 (🏆🏆)
 - NeurIPS 2024: awarded Top Reviewer (Top 8.60%: 1,304 of 15,160 reviewers).
- **ICML**: 2025* (🏆🏆🏆🏆🏆), 2026* (🏆🏆🏆🏆🏆)
 - ICML 2025: awarded Top Reviewer (Top 1.88%: 206 of 10,943 reviewers).
 - ICML 2026: awarded Gold Reviewer (Top 25%: 4,439 of 17,749 reviewers).
- **ICLR**: 2024 (🏆🏆), 2025 (🏆🏆🏆), 2026 (🏆🏆🏆🏆).
- **TMLR** (🏆🏆🏆🏆).

1st GPU server manager of OptiML lab

June 2022 – Feb 2024

- Being involved in installing OptiML lab's very first 5 GPU servers and a storage server
- Allocating GPU nodes to lab members
- Managing errors occurred in the servers

Tutoring Courses for Freshmen at KAIST

Mar 2018 – Dec 2021

- Calculus II (2018–2019; 3 times)
- Introduction to Programming (Fall 2021)

Languages

- English** Professional Proficiency (i.e., sufficient for academic activities)
- Korean** Native proficiency
- Others** Had some introductory courses on French, German, Classical Latin, & Chinese.

Skills

- Programming** Familiar: **Python** 🐍 (**PyTorch**, **NumPy**, **Scikit-learn**, **Jupyter**, **Pandas**, **JAX**, etc.), **MATLAB**.
Novice: C, C++, CUDA, R, HTML/CSS, Scala
- Computer Misc.** Familiar: **TeX** (Overleaf/VSCoDe/MacTex), **Git** 🔒, **Microsoft Office**, **Keynote**
Novice: Adobe products (Lightroom, Premiere Pro, After Effects, Photoshop)
- Music & Hobby** Playing the drums and percussion 🥁. Began learning in 2009.
Joined and performed with the music bands listed below as an amateur drummer:.
- ISHS: *Cha-rang* (2015–2016)
 - KAIST: *Muse KAIST* (2017–2019) → *Carpe Diem* (2019)
 - Club “Music Space”: Team *Woodstone* (2024–2025)
 - Team “*Kira-Kira Yoon(tentative name)*” (2025–current)
- Also, I am a huge music fan, especially of jazz, funk, K-indie, rock, blues, Latin, house, and more.